Code No: 115AN **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, August - 2022
PRINCIPLES OF PROGRAMMING LANGUAGES
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Draw the syntax tree and symbol table for the GCD program.
b) Explain the different categories of languages. [8+7]

2.a) Write and explain the unambiguous grammar for if-then-else.
b) Give the parse tree of the following statement. [8+7]
A = (B+C)*(D/E)

3.a) Describe the design and implementation issues of pointers and reference types.
b) Explain the different types of assignment statements with an example. [8+7]

4.a) Explain the pascal union types with an example.
b) Explain about the type checking and type compatibility. [8+7]

5.a) Explain about the overloaded operators in C++.
b) Explain about the design issues of subprograms. [8+7]

6.a) Explain about the generic sub-programs with an example.
b) Explain about the local referencing environments. [8+7]

7.a) Explain how to handle exceptions in Ada with an example.
b) Explain about the applications of logic programming. [8+7]

8.a) Explain the characteristics of functional programming languages.
b) Write a LISP function fib(n) that computes n^{th} Fibonacci number. [8+7]

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